

TECHBIG SOLUTIONS

Ambattur, Chennai

Oracle Database – Advanced Complete E2E Training Syllabus

Advanced SQL | PL/SQL | DBA | Performance | RAC | Data Guard | Cloud

Program Objective: Build advanced Oracle Database capability covering SQL, PL/SQL, database administration, storage, backup/recovery, performance tuning, security, RAC, ASM, Data Guard, multitenant architecture, migration, monitoring, cloud and production support.

Reference basis: Oracle Database documentation and Oracle learning resources covering Database Administration, SQL/PLSQL, RMAN, Data Guard, RAC, ASM, multitenant architecture, security and cloud database services.

Program	Oracle Database – Advanced Complete E2E
Level	Intermediate → Advanced Oracle Developer / DBA / Consultant
Core Areas	SQL PL/SQL DBA Storage Backup/Recovery Security
Advanced Areas	Performance RAC ASM Data Guard Multitenant OCI
Suggested Duration	140–200 hours, customizable by batch and project depth

1. Oracle Database Architecture

- Oracle Database architecture and major components
- Instance, database, memory and process architecture
- SGA, PGA and background processes
- Control files, data files and online redo logs
- Multitenant architecture: CDB, PDB and services
- Enterprise deployment architecture

2. Oracle Installation & Environment Setup

- Oracle Database installation concepts
- ORACLE_HOME, ORACLE_SID and environment configuration
- Database creation with DBCA
- Listener and client connectivity
- SQL*Plus and SQL Developer
- Development, test, staging and production environments

3. SQL & Advanced Querying

- SELECT, filtering, sorting and aggregation
- Joins and subqueries
- Common Table Expressions (CTEs)
- Set operators
- Analytic/window functions
- Hierarchical queries
- PIVOT/UNPIVOT
- Advanced date, string and numeric functions

4. PL/SQL Programming

- PL/SQL block structure
- Variables, constants and data types
- Control structures
- Loops and collections
- Records and composite types
- Exception handling
- Packages, procedures and functions
- Triggers and reusable PL/SQL design

5. Advanced PL/SQL

- Bulk Collect and FORALL
- Dynamic SQL
- REF CURSORS
- Object types overview
- Pipelined functions
- Autonomous transactions
- Dependency management
- PL/SQL performance and secure coding

6. Schema & Database Object Management

- Users, schemas and objects
- Tables, partitions and constraints
- Indexes
- Views and materialized views
- Sequences and synonyms
- Database links
- Object dependencies and metadata

7. Storage & Tablespace Management

- Tablespaces and data files
- Locally managed tablespaces
- Temporary and undo tablespaces
- Bigfile tablespaces
- Automatic storage management concepts
- Space monitoring
- Segment and extent management

8. Undo & Transaction Management

- Transaction concepts and ACID
- Undo architecture
- Undo retention
- Read consistency
- Commit and rollback
- Savepoints

- ORA-01555 troubleshooting concepts
- Transaction monitoring

9. Redo, Control Files & Recovery Architecture

- Redo generation and redo logs
- Log groups and members
- Control-file management
- Checkpoint concepts
- Archive logging
- ARCH process
- Recovery architecture and failure scenarios

10. Oracle Backup & Recovery with RMAN

- RMAN architecture
- Full, incremental and cumulative backups
- Backup sets and image copies
- Control-file and SPFILE backups
- Restore and recovery
- Point-in-time recovery
- Block media recovery
- RMAN catalog concepts
- Backup validation and testing

11. Data Guard – High Availability & DR

- Physical and logical standby concepts
- Primary/standby architecture
- Redo transport and apply
- Protection modes
- Switchover and failover
- Data Guard Broker
- Fast-start failover concepts
- Monitoring and troubleshooting

12. Real Application Clusters (RAC)

- RAC architecture
- Clusterware and voting/OCR concepts
- Shared storage
- Services and workload distribution
- Instance management
- RAC monitoring
- RAC backup/recovery considerations
- High-availability design

13. Oracle ASM

- ASM architecture

- Disk groups and redundancy
- Failure groups
- ASM disk management
- Rebalancing
- ASM with RAC
- Capacity and performance monitoring

14. Performance Tuning Fundamentals

- Performance-tuning methodology
- CPU, memory, I/O and wait analysis
- Session and workload analysis
- AWR and ASH concepts
- ADDM overview
- SQL performance diagnosis
- Baseline concepts

15. SQL Tuning & Execution Plans

- EXPLAIN PLAN
- DBMS_XPLAN
- Optimizer fundamentals
- Cost-based optimizer
- Statistics and histograms
- SQL profiles and plan management
- Bind variables and cursor sharing
- SQL plan stability

16. Indexing & Partitioning

- B-tree and bitmap indexes
- Composite and function-based indexes
- Index usage and maintenance
- Range, list, hash and interval partitioning
- Partition pruning
- Local vs global indexes
- Partition-wise operations

17. Oracle Optimizer & Statistics

- Optimizer parameters
- Object and system statistics
- Automatic statistics collection
- Histograms
- Cardinality estimation
- Adaptive optimization concepts
- Plan regressions and SQL plan management

18. Security & User Administration

- Users, roles and privileges
- System and object privileges
- Profiles and password policies
- Proxy users
- Role-based security
- Auditing concepts
- Least privilege and secure administration

19. Advanced Oracle Security

- Transparent Data Encryption overview
- Keystore/wallet concepts
- Network encryption concepts
- Virtual Private Database overview
- Fine-Grained Auditing overview
- Database Vault overview
- Sensitive-data protection and compliance

20. Networking & Oracle Net Services

- Listener architecture
- tnsnames.ora and sqlnet.ora
- Service names and connect descriptors
- Dedicated/shared server concepts
- Connection troubleshooting
- Network encryption and authentication
- Client connectivity best practices

21. Oracle Data Pump & Data Migration

- expdp and impdp
- Schemas, tables and full database exports
- Metadata-only and data-only operations
- Remapping schemas/tablespaces
- Parallel import/export
- Migration validation and reconciliation
- Transportable tablespaces concepts

22. GoldenGate & Replication Overview

- Oracle GoldenGate architecture
- Extract, trail and Replicat concepts
- Initial load and change-data capture
- Replication topologies
- Monitoring and lag
- Conflict/error handling
- High-availability and integration use cases

23. Multitenant CDB/PDB Administration

- Container database architecture
- Creating and managing PDBs
- Pluggable database services
- PDB cloning and unplug/plug concepts
- Common vs local users
- PDB backup and recovery
- Resource management and monitoring

24. Oracle Scheduler & Automation

- DBMS_SCHEDULER
- Jobs, programs and schedules
- Job chains and dependencies
- Credentials and external jobs
- Monitoring job runs
- Failure handling
- Automated maintenance and operational tasks

25. Monitoring & Diagnostic Tools

- Dynamic performance views
- Data dictionary views
- Alert log
- Trace files and ADR
- ADRCI
- Session monitoring
- Blocking and deadlock analysis
- Enterprise Manager concepts

26. Locking, Blocking & Deadlocks

- Oracle locking mechanisms
- Row-level and table-level locks
- Blocking-session identification
- Deadlock diagnosis
- Transaction design
- Resolution and prevention strategies

27. Patching, Upgrades & Lifecycle Management

- Patch concepts and release updates
- OPatch overview
- Database upgrade planning
- Pre-upgrade checks
- Upgrade testing and rollback planning
- Invalid objects and post-upgrade validation
- Application compatibility

28. Oracle Cloud & Autonomous Database Overview

- Oracle Cloud Infrastructure (OCI) database services
- Oracle Database Cloud Service concepts
- Autonomous Database overview
- Cloud connectivity and security
- Migration to OCI
- Backup/DR in cloud
- Hybrid Oracle environments

29. Enterprise Monitoring, Capacity & Governance

- Capacity planning
- CPU, memory, storage and growth forecasting
- Performance baselines
- Operational dashboards
- Patch/change governance
- Documentation and runbooks
- Production standards and SLA support

30. Real-Time E2E Oracle DBA Projects

- Enterprise OLTP database installation and configuration
- RMAN backup, restore and point-in-time recovery project
- Data Guard primary/standby implementation and failover testing
- RAC and ASM high-availability architecture project
- Performance tuning using AWR/ASH and execution plans
- Advanced SQL and PL/SQL application development
- Data Pump migration between environments
- Multitenant CDB/PDB implementation and administration
- Oracle security, auditing and access-control project
- Production Oracle DBA support, monitoring and disaster-recovery project

31. Production Support & Troubleshooting

- Database startup/shutdown issues
- Listener and connectivity failures
- Tablespace and storage problems
- Blocking/deadlocks
- High CPU/I/O incidents
- Slow SQL and plan regressions
- RMAN backup/recovery failures
- Data Guard lag and synchronization issues
- RAC/ASM incidents
- ORA-error diagnosis, root-cause analysis and incident management

32. Interview & Certification Preparation

- Oracle SQL and PL/SQL interview questions
- Advanced DBA scenario questions

- Performance-tuning case studies
- RMAN and recovery scenarios
- Data Guard and RAC questions
- ASM and storage scenarios
- Security and multitenant questions
- Real-time project explanation
- Mock practical assessment and Oracle certification preparation

Recommended Hands-on Deliverables

- Install and configure an Oracle database with listener, users, schemas, tablespaces and production-style settings.
- Develop advanced SQL and PL/SQL using analytic functions, CTEs, packages, bulk processing and exception handling.
- Implement RMAN backup/recovery including restore validation and point-in-time recovery.
- Configure a Data Guard environment and perform switchover/failover testing.
- Analyze and tune slow SQL using AWR/ASH, execution plans, statistics and indexing.
- Design an enterprise Oracle HA/DR environment using RAC, ASM and Data Guard concepts.

Training Outcome

Learners completing this program should be able to develop advanced Oracle SQL/PLSQL solutions, administer enterprise Oracle databases, implement backup and recovery, troubleshoot performance and availability problems, manage RAC/Data Guard environments, secure databases and support production Oracle workloads.

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